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Base Station Certificate and Authentication for 5G Radio Control Security

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***Abstract*—Current cellular networking remains vulnerable to fake base stations due to the lack of base station authentication mechanism or even a key to enable authentication. We design and build a base station certificate (certifying the base station's public key and location) and a multi-factor authentication (making use of the certificate and the information transmitted in the online radio control communications) to provide authenticity of the source base station and its radio resource control communications. We advance beyond the state-of-the-art research by introducing greater authentication factors and by using blockchain to deliver the base station digital certificate offline, enabling greater key length/security strength and computational/networking efficiency. The multi-factor authentication at the user equipment involves multiple factors verified through the ledger database, the location sensing, and the cryptographic digital signature verification of the cellular radio control communication (SIB1 broadcasting). We analyze our scheme's security, performance, and the fit to the existing standardized networking protocols. Our work involves the implementation building on X.509 certificate (adapted), smart contract-based blockchain, 5G-standardized radio resource control communications, and software-defined radios. Our analyses show that our scheme effectively defends against more security threats and can enable stronger security, i.e., ECDSA with greater key lengths. Furthermore, our scheme achieves over threefold improvement in computing and energy efficiency on the mobile user equipment compared to previous research.**

***Index Terms*—Base Station Certificate, Base Station Authentication, Fake Base Station, 5G, Blockchain**

I. Introduction

The fake base station has been a well-known security issue in the wireless security community, as described in Section III-A. The fake base station threats include breaching the user privacy against its location and credentials, redirection of the networking (e.g., fake destination and web server), protocol downgrading and manipulation (e.g., de-registration and authentication-and-key-agreement or AKA bypass), and complete control of the availability and wireless link for DoS, among others. The vulnerability stems from the lack of key establishment and authentication of the base stations, which prevents a mobile user from distinguishing between legitimate vs. fake base stations in the radio control connection. In the current practice of 5G, there is no authentication mechanism or even key for the user equipment to authenticate or verify the integrity of the radio control communications.

There have been recent proposals in research [1]–[4] and standardization/development [5] to introduce and use the base station's public key. In 3GPP standardization, such proposals are at the early stages of establishing the requirements for the base station keys [5]. The lack of key disables the authentication and the integrity protection of the control communication exchanged between the user equipment and base station; such control communication resolves the medium access control (MAC) and radio resource control (RRC) to set up the wireless channel before involving the backend core network.

In contrast to the recent research that focuses on base station public-key/certificate to provide only radio control message authenticity, we introduce and design a base station digital public-key certificate to authenticate both the source base station and the message. Our base station certificate includes additional information to provide such authenticity. We design and build our scheme across the base station certificate construction, delivery, and use for multi-factor authentication.

Our base station certificate *construction*, builds on the popular and standardized X.509 certificate, includes the base station's geographical location data, and binds the location with its public key and unique identifier (Cell ID). The base station location information is static, and its addition in the certificate specifically enables additional geographical-sensing-based verification and defense against wormhole threats.

Our base station certificate *delivery* shares the certificate with user equipment to verify the base station's authenticity in radio control communication during cellular access. We use blockchain for such delivery that provides a transparent database of the registered base stations and enables the *offline* delivery of the certificates so that the user equipment can use them during *online*. We describe offline vs. online in Section II-B. The offline delivery provides energy/computing efficiency to the user equipment. It also enables the application of the cryptographic cipher with greater key length/security than the State-of-the-Art research (which delivers the key online) due to the 3GPP-standardized protocol limiting the control message size (i.e., the base station's SIB1 message is limited to 372 B as described in Section II-A).

We then use the offline-delivered base station certificates to enable the *multi-factor authentication* of the base station and its online radio control communication. Our multi-factor authentication verifies not only the authenticity and integrity of the radio control messages (as the State-of-the-Art research does) but also verifies the identity, location, and timestamp of the source base station. The user equipment uses the multiple factors of base station identity, location, timestamp, and public key for sequential authentication of the base station. These factors are verified and cross-checked with multiple

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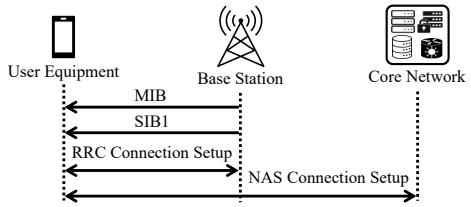

Fig. 1: Control connection setup among user equipment, base station, and core network.

sources, from the stored ledger to the location sensing to the public-key verification. First, the user equipment verifies the identity of the base station using its locally stored blockchain ledger to prove the legitimacy of the base station identity. Second, it verifies the location of the base station from its registered location information, stored in the local ledger. Third, it verifies the freshness of the radio control message using the timestamp. Fourth, the user equipment verifies the message authenticity of the digitally signed radio control message using the base station's public key. The subject base station is authenticated if all the factors are authenticated and verified. The identity and location authenticate the source base station, and the timestamp and public key authenticate the radio control messages.

To highlight our contributions, we design our base station certificate scheme spanning across construction, delivery, and use for multi-factor authentication (§V); both the certificate construction of the information fields and the offline blockchain-based delivery enables the multi-factor authentication and advances its security and performance beyond the existing practice and research. We implement our scheme using srsRAN and Ethereum smart contract for software, and SDR and different computing platforms for hardware (§VI). Our implementation experimentally validates our scheme for proof-of-concept and for empirical security and performance analyses (§VII). We compare our scheme with the State-of-the-Art research in design and approaches (§III-C) and in empirical validation (§VII-A,§VII-C2) to show that our scheme defends greater threats including the wormhole threat, enables digital signature with longer key length, and the time/energy are more than three times efficient at the user equipment.

II. Background

We describe the 5G cellular networking protocol, offline vs. online communications, and the standard cryptographic techniques of public-key digital signature and certificate to describe our scheme in Section V, including the notations and variables for the digital signature.

A. Radio Control in 5G Cellular Network: SIB and RRC

5G cellular network comprises of: user equipment, base station, and core network. The base station acts as a bridge gateway between the wireless and the wired communications, i.e., makes a physical connection with the user equipment through wireless channels and with the core network using wired networking (including routers and switches) for the user equipment to access the remote data and services beyond the immediate base station. For connectivity, user equipment establishes a radio control connection called Radio Resource Control (RRC) with the base station first and then sets up Non-Access Stratum (NAS) layer connection with the core network, as shown in Figure 1. Our work focuses solely on the RRC communications between the user equipment and the base station.

Numerous base stations provide coverage across geographical regions. Each base station provides connectivity in multiple cells (typically three or more, depending on the radio resources, including the MIMO antennas). A base station can be uniquely identified by its cell ID, which is unique within a cellular service provider's network. We thus use the cell ID to identify the base station. A base station can have multiple cell IDs, and each cell ID requires a public key in our scheme.

The base station publicly broadcasts and advertises its presence to the nearby user equipment using the System Information (SI) messages of Master Information Block (MIB) and System Information Blocks (SIBs). The cell-specific System Information Block Type 1 (SIB1) message includes the system parameters (e.g., the cell ID, network ID, and channel frequency information) for the user equipment to bootstrap the radio connection setup, and it is required for the RRC and the following digital communications. During the initial radio connection, the user equipment scans for these broadcast messages and chooses a suitable base station cell to set up the RRC connection based on the signal strength of the SI messages. Because SIB1 is regularly and periodically broadcast, we use SIB1 for the base station authentication in our scheme. In 3GPP standardization, the physical layer limits the maximum packet size of the SIB1 message up to 372 B [6].

Figure 1 shows, the base station broadcasts the regular, periodic MIB and SIB1 messages announcing its presence, including the radio-channel parameters. After receiving these broadcast messages, the user equipment initiates an RRC connection by sending a unicast communication and then completes RRC connection setup with the base station. After a successful RRC connection, the user equipment proceeds to set up a NAS connection with the core network. In this work, we omit the discussion about communication between the base station and the core network, assuming it follows standard protocols.

B. Offline vs. Online

We define online to be during the communication and connectivity provision when the user equipment is in contact with the base station. The *online* operations occur when the user equipment wants to use the cellular service and connectivity, and begins establishing the RRC communication with the base station (in which the user equipment responds to the SIB1 to establish the radio connection).

The *offline* operations occur before the *online* operations, i.e., before the user equipment initiates RRC communications with the base station. The offline operations include the user equipment registration, including the Universal Subscriber Identity Module (USIM) setup and any previous communi-

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cation sessions that the user equipment went through. USIM includes the core network's public key by having it coded in during the user equipment registration. Using such a public key, the core network and user equipment perform mutual authentication and establish the security parameters through Authentication and Key Agreement (AKA) protocol [7].

Because we define offline to be any previous communications before the online, when the user equipment is communicating for payload and encounters the base station, the offline costs in computing, networking, and energy are significantly cheaper than the corresponding online costs. The online costs delay the communication connection in real time, while the offline costs are in advance of the communication connection.

C. Digital Signature and Certificate Variable Notations

Public-key cryptography is widely used for digital signatures, ensuring sender and message authenticity, and for public-key infrastructure (PKI), which issues digital certificates to bind public keys to identities, serving as a root of trust. For digital signature, the subject generates a public-private key-pair (K is the public key while k is the private key) and digitally signs a message m using its private key k to generate the digital signature $S_k(m)$, which involves the public-key cipher algorithm using the private key k on the hash of the message m. The verifier receives the message m and the signature $S_k(m)$, as two inputs for the verification $V_K(m, S_k(m))$. The verification i) computes the hash of m and ii) applies the public-key cipher algorithm using the public key K on $S_k(m)$. If the two results in the same, then the signature is verified; if not, the verification fails. The key inputs for signing S and verification V are in subscripts, while the message inputs are in parentheses. In this research, base station is the subject and user equipment is the verifier and our goal is to authenticate the base station using its public key.

III. Related Work and Our Contribution

This section discusses the fake base station attacks, describes SOTA as a defense in base station public key and authentication, and compare our scheme with SOTA.

A. Fake Base Station Attacks

A fake base station attack uses the RRC communication between the base station and the user equipment. Because there is no base station authentication in current mobile networking, including the most recent 5G, a fake base station can have the mobile user connect to itself by advertising higher channel quality and signal strength [8]–[10]. Once a gullible mobile user connects to a fake base station controlled by an attacker, the attacker can send false public warning alerts [11], [12], launch DNS redirection attack [13], bypass the 5G authentication and key agreement (AKA), and/or deregister the user from the core network [14], downgrade the user to 2G/3G network [9], [15], drain the user's battery [16], [17], launch SMS phishing [18], [19], and track the mobile user's location (violating the user privacy, critical in mobile computing) [9], [17], [20]. The base station has dominant control over the user's connectivity availability and can disrupt it via denial-of-service [9], [17], [20], [21].

Our work is motivated by these threats discovered by previous research. We authenticate the base station to enable the mobile user to distinguish between a legitimate vs. fake base station. It is a preventive defense measure, unlike other defense research for passive detection (e.g., [21]–[23]) or avoidance-based recovery after the attack (e.g., [24]).

B. State of the Art Research (SOTA) Defense

We compare our scheme with the State-of-the-Art research (SOTA), for which we combine and synthesize the research works, making contributions in the certificate construction and use in base station communication. Recent research [1]–[4] proposed using the RRC bootstrapping communications to share the base station's public key and authenticate the base station via digital signature, while [25] also authenticated the base station during handover using authentication tokens. However, the recent research in [1]–[4], [26], [27] are particularly relevant and similar to our work in that they provide systems and implementation-based study, build on 4G/5G networking, use RRC SIB1 message, and build on the standard cryptographic techniques based on digital signature and public-key-infrastructure (PKI)-based authentication.

For certificate/key delivery, Hussain et al. [2] and Ross et al. [3] use the preinstalled public key of the core network in the USIM as the initial anchoring point to generate the certificate chain, and the mobile user verifies the digital signatures of the intermediary nodes (e.g., MME/AMF) and base station through the certificate chain. Ross et al. [3] introduce an additional RRC messages to deliver the certificate chain while others fit the optimized certificate within a byte-size-limited RRC SIB1 message. Building on such key delivery, SOTA [1]–[4] digitally sign the base station's RRC SIB1 message. In these schemes, the user equipment verifies three digital signatures (core network, MME/AMF, SIB1) to authenticate the RRC communications that induce computational load at the user equipment (e.g., foregoing certificate chaining [2], [3] or multi-party computation [1], [4]).

C. Our Contributions Beyond SOTA

We compare our work with SOTA based on the synthesis of the collection of research works described in Section III-B. Our work partly builds on these SOTA works in that we also rely on the core network's public key delivery via USIM and the digital signature encoding in the RRC SIB1. However, our work enables greater security and efficiency than SOTA, which itself outperforms the individual research works.

Our system outperforms SOTA in four key security and performance advantages by synergistically co-designing certificate construction, blockchain-based offline delivery, and multi-factor authentication. First, our scheme reduces the number of signature verifications by directly sharing the base station's public key, while the previous research requires three digital signature verifications (for the core network, MME/AMF, and SIB1/base station). Second, the base station certificate includes the base station's geographical location, enabling the user equipment to verify the location authenticity of the base station. The location verification detects the wormhole threats

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while SOTA remains vulnerable. Third, our scheme delivers the base station's certificate including public key, ID, and the location offline via blockchain, while the previous research delivers it along with byte-limited SIB1 online, i.e., appends the public key in SIB1. Our scheme yields the following advantages: provides a more robust delivery of such offline communications, enables ECDSA digital signature algorithms of greater security and key length, reduces the computation and battery power consumption at the user equipment, and compatible with existing 5G standardization. Fourth, the multi-factor authentication uses multiple orthogonal channels to verify the factors presented by the subject base station: the ledger serving as the registered base station list, the real-time location sensing, and the cryptographic signature verification. An attacker needs to compromise all of these channels to break the base station authentication and to inject control communications to get accepted, reducing the threat feasibility. We empirically validate these design goals and advantages.

IV. Threat Model

We consider an attacker acting as a fake base station. The attacker is equipped with radio hardware, well-versed in telecommunications protocol, and within the transmission range of the victim user equipment. The attacker can inject communications and modify or generate the relevant control communication fields, such as ID, public key, or location. The attacker can also listen and capture the communications between the legitimate base station and the user to use that to enable more sophisticated threats. For example, as one of the more advanced threats, we consider a wormhole threat where an attacker uses the captured information from one cell location to craft the communication at another cell location.

Such a threat model based on a radio-equipped attacker is of high risk (high threat feasibility) because it does not require the system compromise of the victim user equipment or the digital credential compromise of the telecommunications service provider infrastructure, such as the core network's public key. It is also of high risk because the base station's control communications are inherently broadcasting, and the current practice has no key or authentication protections. Our threat model aligns and is comparable with that used by other previous research works, described in detail in Section III-B.

Our threat model covers threats in 5G RRC communications of three types: new base station threat, base station spoofing, and control communication injection threat. In a new base station threat, the attacker installs a new fake base station with its own identity (e.g. Cell ID), whereas in base station spoofing, the attacker imitates a legitimate base station's identity. The most sophisticated attack that an attacker can perform is a wormhole attack, which is a form of control communication injection. The attacker can launch a *wormhole* attack by listening to SI messages from a legitimate base station at one location and transmitting them to another location using a fake base station, where the transmission is almost instantaneous, i.e., enough to thwart timing- and freshness-based mechanisms.

Version Number	Serial Number	Issuer Name	Validity Period	Subject Name	Public Key, ***K***	Signature Algorithm	Cell ID ***ID***	Location ***L***	Certificate Signature

Fig. 2: Our base station certificate information fields with the new additions are highlighted in blue. This corresponds to the individual record of a registered base station in the base station certificate database $\mathcal{C}$.

V. Our scheme

We design our scheme to defend against the threats described in Section IV. The base station certificate construction and its offline delivery enable the multi-factor authentication at the user equipment, and such authentication provides defense against threats. In our scheme, the construction is offline; the delivery is offline (while the SOTA research is online as described in Section III-B); and the certificate used for multi-factor authentication is online.

A. Our Scheme Overview in Offline and Online

Our scheme consists of two phases: offline (base station certificate construction and delivery) and online (multi-factor base station authentication). In the offline certificate construction and delivery phase, the base station binds its cell ID (ID) and the geographical location (L) with its public key (K) to construct its certificate. The core network validates and signs the certificate and publishes the certificate via the blockchain ledger. The offline blockchain communication delivers base station certificates to the user equipment at any time before the radio connection with the base station, e.g., in any of the previous communications by the user equipment.

During the online multi-factor authentication phase, the base station digitally signs the first RRC message, i.e., SIB1, and the user equipment receives and verifies the SIB1 message using the ID, location, and public key of the base station that it gets from the offline blockchain.

In offline, we deliver the digital certificate database of the base stations $\mathcal{C} = \{C_i\}_i$, where each record $C_i = \{ID_i, K_i, L_i, ...\}$ is for the registered base station index i and i iterates across the base stations for each row. For example, the first record/row corresponds to the first registered base station $i = 1$, i.e., $C_1 = \{ID_1, K_1, L_1, ...\}$. Each certificate C_i can include other fields as indicated in Figure 2, but ID_i, K_i, L_i are the most relevant to our scheme. In online, we deliver cell ID (included in SIB1), nonce (n), and SIB1 generation timestamp (t) appended into the SIB1 message of the base station i. The formation from offline and online gets used for authenticating the base station i during multi-factor authentication. We describe such information deliveries in greater detail in the next sections.

B. Base Station Certificate Construction (Offline)

Our base station certificate construction builds on the X.509 standardized certificate, which is popularly used for securing the Internet and web. We adapt its construction specifically for base stations and to enable greater security objectives. Figure 2 illustrates the data fields within our base station certificate,

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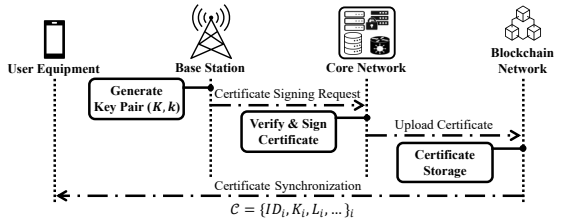

Fig. 3: Offline certificate construction and delivery. The certificate construction involves the base station and core network (the inner middle entities), while the delivery further involves the user equipment (left) and blockchain (right).

where our additions beyond X.509 are highlighted in blue. We use the base station's cell ID (ID) in the ID field of the certificate to uniquely identify each base station.

We add the geographical location (L) of the stationary terrestrial base station in the certificate. L is a two-dimensional object containing the longitude and latitude of the base station i.e., L=(L_1,L_2), where L_1, L_2 are the longitude and latitude coordinate respectively. Base station's location is public information, unlike the user equipment's location and behavior, which require privacy, and is thus appropriate for widespread delivery. For example, in the US, the FCC keeps track of the registered base stations and publishes their cell IDs and the geographical locations [28]. We use this data in our prototype implementation for simulating the real-world base station deployments.

While Figure 3 illustrates our scheme protocol for both the certificate construction and delivery, this section focuses on the certificate construction between the core network and the base station. The certificate construction involves the base station (for public-private key pair generation) and the core network (for signing the certificate), whose computing and communications are drawn in Figure 3 between the middle two entities. The figure shows, the certificate construction process begins with the base station. During the installation of the new base station or when the current digital certificate expires, the base station generates its public-private key pair (K, k) for each cell. Then the base station constructs its X.509 certificate (shown in Figure 2) and sends a certificate signing request to the core network. The base station creates separate certificate signing requests for each of its cells and sends them to the core network for signing, generating one certificate per ID. Upon receiving the signing request, the core network validates the information and digitally signs to generate the certificate signature (in the far-right field in Figure 2). Section V-C1 describes the delivery part involving blockchain and others.

C. Certificate Delivery Using Blockchain (Offline)

1) Blockchain Enabler, Appropriateness, and Effectiveness: Blockchain enables the secure and offline delivery of the base stations' certificates. Blockchain is appropriate for such delivery because, we store the public information of the base station in the blockchain but avoid the privacy-sensitive or user-equipment-specific information.

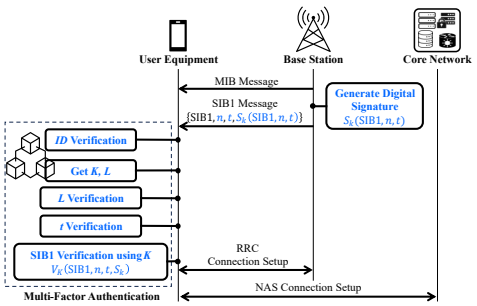

Fig. 4: Online multi-factor authentication by user equipment, including the user-side computing for sequential verifications (from top to bottom on the left of the diagram). The blue color indicates our addition in 5G RRC.

Because the service provider infrastructure (core network and base stations) are public entities that do not require anonymity or privacy and because the public keys should be publicly known for the security of the public-key digital signature, we use a permissioned but public blockchain where the blockchain ledger gets written only by the core network while it can be read/stored/accessed by all the entities in the cellular networking. Therefore, our blockchain forgoes the heavily distributed consensus protocol, e.g., computation-heavy proof-of-work (PoW), to determine which data gets uploaded and by whom. (Technically, our blockchain implementation in Section VII-B uses PoW consensus but with a difficulty of 0, i.e., all hash function computation for mining is valid.) Also, in contrast to the cryptocurrency blockchain, we do not have the incentivization or competition issues because the service provider infrastructure and its beneficiaries/clients are all aligned in their incentives to provide connectivity and secure cellular connections against the rogue base stations. Our use of the blockchain is thus substantially simpler and cheaper than that used in the permissionless applications, such as those for cryptocurrencies and financial transactions.

Blockchain is effective for our base station certificate delivery. Blockchain provides message authentication integrity anchored by the security-robust cryptographic hash function and public-key digital signature. Blockchain provides transparent access to the database/ledger thanks to its distributed storage property; in our scheme, the database includes the registered base stations, and blockchain transparency enables accountability. Blockchain also provides automatic synchronization; in our scheme, it can provide the updated base station certificates offline in ad hoc connections, using other non-cellular technology, or from the previous base station connections. We experimentally validate and analyze the appropriateness and effectiveness of blockchain for offline base station certificate delivery in Section VII-B, while omitting the efficiency validations.

2) Certificate Delivery Using Blockchain: The core network (after responding to the base station's certificate request as described in Section V-B) uploads the base station certificates to the blockchain, as shown in Figure 3. The blockchain

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ledger includes the digital certificates that are signed by the core network and whose subjects are the base stations (as well as the core network itself in the genesis block to enable the writing/uploading of the certificate). The blockchain ledger serves as the database ($\mathcal{C}$) of registered base stations for the user equipment. The integrity of such records (i.e., only the core network can write and insert) relies on blockchain technology, as described in Section V-C1.

The base station certificate ($\mathcal{C}$) gets delivered offline via blockchain. The user equipment uses the core network's public key stored within its USIM card to verify the authenticity of $\mathcal{C}$ and, upon verification, store $\mathcal{C}$ in its local storage. This process occurs offline, which is any time before the user becomes active and connects to the cellular base station as described in Section II-B. The user synchronizes the ledger ($\mathcal{C}$) using non-cellular connectivity (e.g. as described in Section VI) only during the initial subscription setup. Subsequent updates to $\mathcal{C}$ can be synchronized using the previously established radio connection, e.g., via the last connected base station.

D. Multi-Factor Authentication (Online)

In online, when the user equipment connects to a base station using RRC protocol, it implements *multi-factor authentication* to authenticate the base station using multiple factors. The multi-factor authentication involves the sequential verification of the subject base station transmission using multiple factors (ID, L, t, SIB1) received from multiple sources (offline, online, local sensing). We show the multi-factor authentication within the 5G RRC communication in Figure 4 and highlight our scheme in blue font color, while the existing RRC protocol is shown in black.

1) Signature Generation at Base Station: Our scheme uses the essential and periodic broadcast message, SIB1, for the base station message authentication as described in Section II-A. As shown in Figure 4, before broadcasting SIB1 message, the base station generates a digital signature S_k(SIB1,n,t) using its private key (k). We use the SIB1 signature generation timestamp (t) and a nonce (n) for a replay attack protection. The base station appends the digital signature and other parameters to the SIB1 message and broadcasts it to the user equipment. The final SIB1 transmission appends the following: {SIB1,n,t,S_k(SIB1,n,t)}.

2) The Factors and The Verification Sources: In offline, the user equipment receives the base station's certificate information $\mathcal{C} = \{ID_i, K_i, L_i\}_i$, which is stored in its local ledger as described in Section V-C2. In online, i.e., when the user equipment wants cellular connectivity, the user equipment acquires SIB1, n, t, and $S_k(\text{SIB1}, n, t)$ from the periodic base station SIB1 transmission. The user equipment also locally senses its location $\bar{L}$ and measures the SIB1 reception time $\bar{t}$ to verify the provided L (in offline) and t (in online). The user equipment controls the parameters, τ_d and τ_t, for the threshold-based verifications of L and t.

3) Multi-Factor Authentication at User Equipment: We use the factors and the multiple sources described in Section V-D for the sequential authentication as depicted in Figure 4.

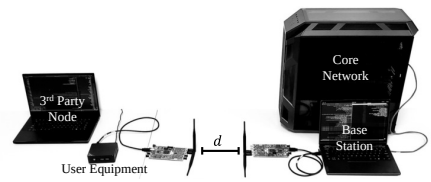

Fig. 5: Experimental setup with the user equipment, base station, core network, and a 3rd party node. USRP B210 SDR is used for 5G communications. The 3rd party node communicates the blockchain via Wi-Fi or Bluetooth.

In ID verification, the user equipment extracts the Cell ID (as described in Section II-A) from the SIB1 message and performs an ID lookup within $\mathcal{C} = \{ID_i, K_i, L_i\}_i$. If the cell ID matches an ID from $\mathcal{C}$, then the ID of the base station is verified. After ID verification, the user equipment accesses the location L, and the public key K of the base station from $\mathcal{C}$ for the location and digital signature verification.

L and t verifications are threshold-based comparisons with thresholds τ_d and τ_t respectively. In L verification, the user equipment uses the distance threshold τ_d to cross-check if the distance between L and $\bar{L}$ is less than τ_d. In t verification, the user equipment uses the time threshold τ_t to cross-check if the time difference between the SIB1 reception time and SIB generation time is less than the threshold, i.e., if ($\bar{t} - t \leq \tau_t$). User equipment can dynamically select τ_t and τ_d based on factors like signal power, SNR, timing advance, and nearby base stations. Formal tuning of these thresholds is not covered here and is proposed as a direction for future research.

Finally, the user equipment verifies the SIB1 message by authenticating the digital signature using base station public key K, accessed from $\mathcal{C}$ after ID verification. It receives the SIB1 message with the digital signature $V_K(S_k$(SIB1,n,t)) during online SIB1 communication and verifies the signature to validate the authenticity and integrity of the SIB1 message using K. The user equipment authenticates the subject base station if all the factors are authenticated and then proceeds with RRC connection with the base station.

VI. Implementation and Experimental Setup

Because we implement both the offline blockchain-based key delivery and its use in the online RRC communications, our implementations include both local (direct communication link) and remote setups. We also successfully implement the fake base station threats described in Section IV against an off-the-shelf phone (victim user equipment) as proof-of-concepts. The technical report version of this paper [29] presents these threat results. Our threat demonstrations corroborate the previous research on fake base stations described in Section III-A.

Entities and Hardware: We simulate the different nodes involved in 5G RRC and blockchain in our scheme. These nodes are User Equipment (Intel 12th Gen N95 mini PC), Base Station (AMD Ryzen 7 5700U), Core Network (AMD Ryzen Threadripper 3960x), and 3rd Party Node (Intel Core i7-10750H) as depicted in Figure 5. The 3rd Party Node transmits

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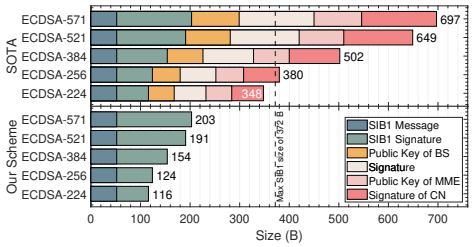
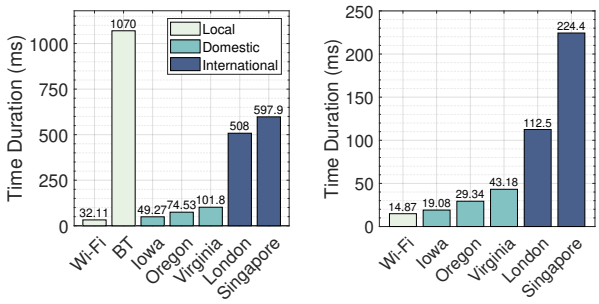


Fig. 6: SIB1 packet size comparison between our scheme and SOTA with different signature algorithms. The vertical line shows the SIB1 size limit of 372 Bytes.



(a) Certificate delivery time. (b) Transmission delay in RTT.

Fig. 7: Offline blockchain-based certificate delivery performances when varying the certificate database source locations (Local vs. Cloud) and the communication protocols (Wi-Fi vs. Bluetooth).

the blockchain communication to the user equipment and represents anybody it communicates through offline, for example, another user equipment or the core network. We use two Ettus USRP B210 SDRs for frontend radio communications between the user equipment and the base station.

Offline (Blockchain): We use Go Ethereum local private blockchain (i.e., independent of the Ethereum cryptocurrency Mainnet) and Solidity smart contract for the blockchain implementation. We locally store the ledger using Redis caching to make the real-time online operation more efficient, as Ethereum creates an unnecessary new transaction for the ledger read in our scheme. We implement offline blockchain communications using diverse communication channels in both local ad hoc and remote infrastructure-based environments. We vary the communication protocols to demonstrate the diverse and robust ways that the blockchain can deliver the base station's public key c.f., Bitcoin even gets delivered via unconventional satellite networking [30]. For the ad hoc local network, the 3rd party node and the user equipment connect to Wi-Fi or Bluetooth to distribute the blockchain ledger. For the remote Internet-infrastructure-based networking, the remote server (Google Cloud servers in domestic and international locations) connects to our user equipment via TCP/IP for the blockchain/key delivery.

Online (RRC): For the online implementation, we incorporate our scheme in srsRAN, an open-source 4G/5G prototyping software. We modify the RRC protocol to implement our scheme as shown in Figure 4. We use OpenSSL for the digital signature and X.509 certificates, whose operations are incorporated in both the online RRC and offline blockchain. In our experimentation, we also vary three different wireless channel conditions between the user equipment and base station to demonstrate a real-world deployment scenario. We present these results and greater details in [29].

VII. Experimental Analyses

A. Security Analyses of the Scheme

We compare our scheme's security with SOTA and empirically verify its effectiveness against threats outlined in Section IV.

1) Enables Greater Cryptographic Strength: Our scheme, utilizing the offline delivery of the certificate and public key, enables more efficient communications, i.e., fewer bits to transfer in the RRC SIB message. Figure 6 shows, our scheme can support all the ECDSA ciphers with varying key lengths and security strengths, while the SIB1 size limit of 372 B (as described in Section II-A) prohibits the use of ECDSA with greater keys beyond ECDSA-224 in SOTA. In addition to theoretically analyzing it based on our understanding of the RRC SIB1 protocol and ECDSA, we verified this in our prototype implementation. While this section focuses on the increased cryptographic security strength against brute-force attackers, we also experimentally validate and analyze the reduced computation and energy use because of the communication data efficiency in Section VII-C.

2) Effective Against Fake Base Station Threats: Our scheme prevents new base station attacks using ID verification. The attack fails because the user equipment verifies the base station cell ID with the registered base station certificate database stored in a blockchain ledger, $\mathcal{C}$. $\mathcal{C}$ contains only the legitimate base station certificates verified by the core network during offline certificate construction described in Section V-B; therefore, any new base station with a fake cell ID will not be authenticated.

In the base station spoofing attack scenario, the fake base station fails to prove the authenticity of its SIB1 messages because of the digital signature. Given a legitimate base station's key-pair (k, K), the attacker cannot forge a valid digital signature, $S_k(\text{SIB1})$, using its private key, k'. Also, the user equipment invalidates a forged digital signature, $S_{k'}(\text{SIB1})$, using the legitimate base station's public key, K, because our usage of permissioned blockchain prevents the attacker from distributing fabricated public keys.

The L and t verification prevents communication injection attack, more specifically, wormhole and replay attack. In wormhole attack implementation, the L verification fails because distance between L and $\bar{L}$ is greater than τ_d. In the replay attack scenario, t verification fails because of $\bar{t} - t > \tau_t$. Although such threshold-based verification might have a risk of false positives or false negatives in verification, however, we leave such threshold selection as future research scope.

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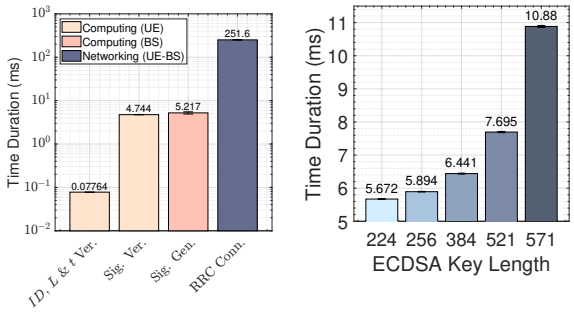
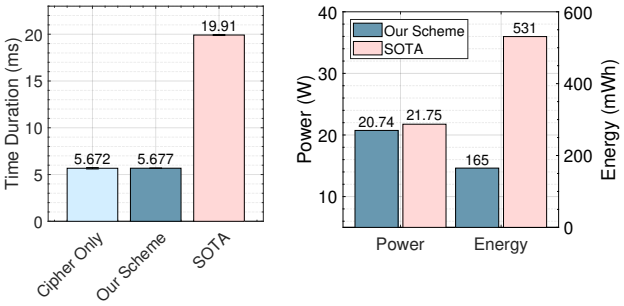


(a) Time durations at the user equipment and base station.

(b) Signature verification times varying ECDSA.

Fig. 8: Our scheme's online performance.



(a) Signature verification time durations at the user equipment

(b) Power (measured) and energy (calculated) consumption at the user equipment

Fig. 9: Our scheme vs. SOTA comparison.

B. Offline Results for Blockchain Key and Certificate Delivery

We implement the offline blockchain for the base station certificate delivery and run experiments to measure the blockchain size, blockchain scalability requirement (the number of certificates per second required to support the 5G applications), and the certificate delivery time.

1) Blockchain Scalability: In the blockchain use for base station certificate delivery, each new transaction or record gets updated when there is a change in the base station registration, including a new base station construction or a public-key revocation, which is far less frequent than the scalability requirements of other current blockchain implementations and practices such as those for financial transactions. Our scheme requires only 0.0266 certificate/transactions per second, which is three orders of magnitude smaller than the other blockchain applications for cryptocurrency such as Ethereum and Bitcoin.

2) Offline Time Overhead Performance: We use diverse communication channels (Wi-Fi/BT) and sources (local/cloud) to demonstrate the robustness of blockchain-based certificate delivery to the user equipment. Figure 7a shows our experimental results for certificate delivery (34680 B) from the 3rd node and the cloud server using ad hoc and infrastructure networks. The ad hoc network connectivity using Wi-Fi provides the fastest certificate delivery (32.22 ms) from the 3rd party node as compared to other locations and connectivity. The certificate transfer using ad hoc Bluetooth takes the longest time (1070.2 ms), which is natural due to the physical-layer design of Bluetooth providing low data rate/bandwidth. We also measure the certificate delivery from the remote cloud servers, connected to the user via the Internet (TCP/IP) and through the 3rd Party Node as the last networking hop via Wi-Fi. As shown in Figure 7b, the RTT time duration increases as the remote node to initiate the blockchain transmission is located geographically farther. The cloud server in Iowa provides the lowest certificate delivery time (52.23 ms) and the cloud server in Singapore takes the highest time (1060.05 ms or 1.06 s) to deliver the certificate. While these measurements can be high and can reach up to about 1 s, the offline cost in the time overhead is significantly cheaper than the real-time online cost in Section VII-C.

C. Online Results for multi-factor Authentication

We compare our scheme against the SOTA described in Section III-B in the computational time, energy consumption, and the RRC connection setup overheads.

1) Our Scheme's Online Performances: We take a systems approach to measure the performance across the online operations of our scheme. In Figure 8a, we show that ID, L, t verification time (0.0776 ms) and signature verification time (4.744 ms) at the user equipment are much lower than the RRC connection establishment time duration between the user equipment and the base station (251.6 ms). The user equipment computing is only 1.89% compared to the entire RRC networking between the user equipment and base station. Also, we add only a total of 72 B of overhead in the SIB1 message using the ECDSA-224 algorithm (signature - 64 B, nonce - 4 B, and timestamp - 4 B).

Our scheme enables the use of the ECDSA ciphers with greater keys and security strengths, while the use of the ECDSA with greater keys is prohibited in SOTA as described in Section VII-A1. We experimentally verify this and measure the time overheads when varying the ECDSA key lengths in Figure 8b to show that the time overheads increase as the key lengths increase.

2) Our Scheme vs. SOTA: We focus on our scheme implementation using ECDSA-224 when comparing our scheme vs. SOTA because SOTA does not support the other ECDSA digital signature algorithms as discussed in Section VII-A1. Figure 9a compares the computing overhead of the signature verification for our scheme vs. the SOTA. Because the SOTA requires three signature verifications on the user equipment due to certificate chaining, while our scheme only requires one, the signature verification overhead of the SOTA (19.91 ms) is 3.50 times greater than our scheme (5.68 ms). Our scheme's signature verification after incorporating it within srsRAN (5.68 ms) is close to the digital signature verification without the srsRAN incorporation (5.67 ms).

The signature verification efficiency of our scheme compared to the SOTA results in the energy efficiency of our scheme, which is important for battery-equipped and energy-constrained user equipment. We take the power measurements

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in our experiment and then calculate the energy consumption by multiplying the average power consumption and the average time duration for the signature verification. As shown in Figure 9b, the SOTA takes 3.22 times more energy (531.38 mWh versus 165.21 mWh) than our scheme to verify the signatures, because our scheme verifies the digital signature once, while the SOTA verifies three digital signatures to complete the authentication.

VIII. Conclusion

To defend against fake base stations, we construct a base station certificate that includes its ID, public key, and location information, then deliver the certificate from the network to the user equipment using blockchain to enable multi-factor base station authentication at the user equipment. We use offline communication for base station certificate construction and delivery while using online communication for multi-factor base station authentication which involves sequential verifications. Our work advances security and performance over the SOTA to enable authentication using multiple orthogonal channels on the user equipment, to defend against wormhole using the location information, and to enable greater performance and security strengths. Our scheme involving a novel base station certificate and its blockchain-based delivery and use in 5G RRC enables stronger security in ECDSA cipher with greater key length. Our scheme also provides more than three times better performance advantages in computational time and energy consumption by verifying the digital signature only once, unlike SOTA. We experimentally validate the security and performance advantages of our scheme over SOTA by implementing it using open-source software (srsRAN, Ethereum, OpenSSL) and SDR.

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References

[1] A. Singla, R. Behnia, S. R. Hussain, A. Yavuz, and E. Bertino, "Look before you leap: Secure connection bootstrapping for 5g networks to defend against fake base-stations," in *Proceedings of the 2021 ACM Asia Conference on Computer and Communications Security*, 2021.

[2] S. R. Hussain, M. Echeverria, A. Singla, O. Chowdhury, and E. Bertino, "Insecure connection bootstrapping in cellular networks: The root of all evil," in *Proceedings of the 12th Conference on Security and Privacy in Wireless and Mobile Networks*, ser. WiSec '19, 2019.

[3] A. J. Ross, B. Reaves, Y. Nasser, G. Cukierman, and R. P. Jover, "Fixing insecure cellular system information broadcasts for good," in *Proceedings of the 27th International Symposium on Research in Attacks, Intrusions and Defenses*, 2024, pp. 693–708.

[4] Y. Dong, R. Behnia, A. A. Yavuz, and S. R. Hussain, "Securing 5g bootstrapping: A two-layer ibs authentication protocol," *arXiv preprint arXiv:2502.04915*, 2025.

[5] 3GPP. TR 33.809 version 0.8.0 , "Study on 5G security enhancements against false base stations," 2021.

[6] 3GPP. TS 33.331 version 17.2.0 , "5G NR; Radio Resource Control (RRC); Protocol specification," 2022.

[7] 3GPP. TS 33.501 version 16.13.0 , "Security architecture and procedures for 5G System," 2023.

[8] H. Yang, S. Bae, M. Son, H. Kim, S. M. Kim, and Y. Kim, "Hiding in plain signal: Physical signal overshadowing attack on {LTE}," in *28th USENIX Security Symposium (USENIX Security 19)*, 2019, pp. 55–72.

[9] A. Shaik, R. Borgaonkar, N. Asokan, V. Niemi, and J.-P. Seifert, "Practical attacks against privacy and availability in 4g/lte mobile communication systems," 2015.

[10] E. Bitsikas and C. Pöpper, "Don't hand it over: Vulnerabilities in the handover procedure of cellular telecommunications," in *Proceedings of the 37th Annual Computer Security Applications Conference*, 2021.

[11] ——, "You have been warned: Abusing 5g's warning and emergency systems," in *Proceedings of the 38th Annual Computer Security Applications Conference*, 2022, pp. 561–575.

[12] G. Lee, J. Lee, J. Lee, Y. Im, M. Hollingsworth, E. Wustrow, D. Grunwald, and S. Ha, "This is your president speaking: Spoofing alerts in 4g lte networks," in *Proceedings of the 17th Annual International Conference on Mobile Systems, Applications, and Services*, 2019.

[13] D. Rupprecht, K. Kohls, T. Holz, and C. Pöpper, "Breaking lte on layer two," in *2019 IEEE Symposium on Security and Privacy (SP)*. IEEE, 2019, pp. 1121–1136.

[14] H. Kim, J. Lee, E. Lee, and Y. Kim, "Touching the untouchables: Dynamic security analysis of the lte control plane," in *2019 IEEE Symposium on Security and Privacy (SP)*. IEEE, 2019, pp. 1153–1168.

[15] B. Karakoc, N. Fürste, D. Rupprecht, and K. Kohls, "Never let me down again: Bidding-down attacks and mitigations in 5g and 4g," in *Proceedings of the 16th ACM Conference on Security and Privacy in Wireless and Mobile Networks*, ser. WiSec '23, 2023.

[16] A. Shaik, R. Borgaonkar, S. Park, and J.-P. Seifert, "New vulnerabilities in 4g and 5g cellular access network protocols: Exposing device capabilities," in *Proceedings of the 12th Conference on Security and Privacy in Wireless and Mobile Networks*, ser. WiSec '19, 2019.

[17] S. R. Hussain, M. Echeverria, I. Karim, O. Chowdhury, and E. Bertino, "5greasoner: A property-directed security and privacy analysis framework for 5g cellular network protocol," in *Proceedings of the 2019 ACM SIGSAC Conference on Computer and Communications Security*, ser. CCS '19, 2019, p. 669–684.

[18] Y. Zhang, B. Liu, C. Lu, Z. Li, H. Duan, S. Hao, M. Liu, Y. Liu, D. Wang, and Q. Li, "Lies in the air: Characterizing fake-base-station spam ecosystem in china," in *Proceedings of the 2020 ACM SIGSAC Conference on Computer and Communications Security*, 2020.

[19] H. Wen, P. Porras, V. Yegneswaran, and Z. Lin, "Thwarting smartphone sms attacks at the radio interface layer," in *30th Annual Network and Distributed System Security Symposium, NDSS*, 2023.

[20] S. Hussain, O. Chowdhury, S. Mehnaz, and E. Bertino, "Lteinspector: A systematic approach for adversarial testing of 4g lte," in *Network and Distributed Systems Security (NDSS) Symposium 2018*, 2018.

[21] S.-Y. Chang and S. Purification, "Securing cellular availability: The wireless blackhole threat and defense," in *2025 IEEE Conference on Communications and Network Security (CNS) (IEEE CNS 2025)*, Avignon, France, 2025.

[22] Z. Li, W. Wang, C. Wilson, J. Chen, C. Qian, T. Jung, L. Zhang, K. Liu, X. Li, and Y. Liu, "Fbs-radar: Uncovering fake base stations at scale in the wild." in *NDSS*, 2017.

[23] S. Purification, S. Wuthier, J. Kim, J. Kim, and S.-Y. Chang, "Fake base station detection and blacklisting," in *2024 33rd International Conference on Computer Communications and Networks (ICCCN)*, 2024, pp. 1–9.

[24] S. Purification, J. Kim, J. Kim, and S.-Y. Chang, "Fake base station detection and link routing defense," *Electronics*, vol. 13, no. 17, 2024.

[25] A. Lotto, V. Singh, B. Ramasubramanian, A. Brighente, M. Conti, and R. Poovendran, "Baron: Base-station authentication through core network for mobility management in 5g networks," in *Proceedings of the 16th ACM Conference on Security and Privacy in Wireless and Mobile Networks*, ser. WiSec '23, 2023, p. 133–144.

[26] S. Purification and S.-Y. Chang, "Verifiable alerts for 4G/5G public warning system," in *2025 IEEE Conference on Communications and Network Security (CNS) (IEEE CNS 2025)*, Avignon, France, 2025.

[27] Y. Dong, T. Wan, T. Wu, and S. R. Hussain, "Evaluating time-bounded defense against rrc relay in 5g broadcast messages," in *18th ACM Conference on Security and Privacy in Wireless and Mobile Networks*, 2025, pp. 236–241.

[28] Federal Communications Commission, "Cellular tower locations," https://shorturl.at/zGLpm, 2025, [Online; accessed 21-July-2025].

[29] S. Purification, S. Wuthier, J. Kim, I. Kim, and S.-Y. Chang, "Base station certificate and multi-factor authentication for cellular radio control communication security," 2025. [Online]. Available: https://arxiv.org/abs/2504.02133

[30] Blockstream, "Rethinking Trust." https://blockstream.com/satellite/, 2025, [Online; accessed August-2025].

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